

T3 matched BSM $\Delta\mathcal{L}_{\text{EFT}}$ terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from matched BSM EFT terms. Empty cells denote no matched term with that field combination.

Field combinations 1 of 3

d_S	Y_S	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H\eta$
2	$\frac{1}{2}$	1	0	$-\frac{\hbar B_{\mu_1\mu_2}^2}{24\epsilon}$	$+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{\hbar H_{a_1}H_{a_2}\lambda_3^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{\hbar H_{a_1}H_{a_2}\lambda_4^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{\hbar H_{a_1}H_{a_2}\lambda_3\lambda_4H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{\hbar\overline{\lambda_5}H_{a_1}H_{a_2}\lambda_5H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$	$+\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_3^2H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_4^2H_{a_1}^\dagger H_{a_2}^\dagger$ $+\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_3\lambda_4H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{\lambda_5}H_{a_1}H_{a_2}\lambda_5H_{a_1}^\dagger H_{a_2}^\dagger$
2	$\frac{1}{2}$	3	0	$-\frac{\hbar B_{\mu_1\mu_2}^2}{24\epsilon}$	$+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{\hbar H_{a_1}H_{a_2}\lambda_3^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{\hbar H_{a_1}H_{a_2}\lambda_4^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{\hbar H_{a_1}H_{a_2}\lambda_3\lambda_4H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{\hbar\overline{\lambda_5}H_{a_1}H_{a_2}\lambda_5H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$	$-\frac{25\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon}$	$+\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_3^2H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_4^2H_{a_1}^\dagger H_{a_2}^\dagger$ $+\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_3\lambda_4H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{\lambda_5}H_{a_1}H_{a_2}\lambda_5H_{a_1}^\dagger H_{a_2}^\dagger$
3	$\frac{1}{2}$	2	0	$-\frac{\hbar B_{\mu_1\mu_2}^2}{16\epsilon}$	$+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{3\hbar H_{a_1}H_{a_2}\lambda_3^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$	$-\frac{2\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon}$	$+\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_3^2H_{a_1}^\dagger H_{a_2}^\dagger$

Field combinations 2 of 3

d_S	Y_S	d_F	Y_F	N	$W_{\mu\nu}$	$\eta B_{\mu\nu}$	$\eta W_{\mu\nu}$	$H \ell e$
2	$\frac{1}{2}$	1	0			$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $-\frac{1}{8}\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $-\frac{1}{8}\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger$
2	$\frac{1}{2}$	3	0		$-\hbar\log\left(\frac{\mu\text{bar}2}{M_N^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{3\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon}$ $-\frac{3}{32}\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger}{16\epsilon}$ $-\frac{3}{32}\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger$
3	$\frac{1}{2}$	2	0		$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_N^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{16}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{3\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon}$ $-\frac{3}{32}\hbar\overline{y_{N,i_3,r_1}}\overline{\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger}{16\epsilon}$ $-\frac{3}{32}\hbar\overline{y_{N,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger$

Field combinations 3 of 3

[illegible]

$d_S \quad Y_S \quad d_F \quad Y_F \quad H \quad \eta \quad N \quad \ell$

$H \quad \eta \quad N \quad \ell \quad e$

$2 \quad \frac{1}{2} \quad 3 \quad 0$

$$\begin{aligned}
& + \frac{h \overline{y_{N,i_1,r_1}} \overline{y_{N,i_2,r_1}} \overline{\lambda_5} \overline{\ell_{a_3,i_1}}^T C P_L \ell_{a_2,i_2} H_{a_1} H_{a_4} \overline{\ell_{a_1,a_2}} \overline{\ell_{a_3,a_4}} M_N^3}{8 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{h \log \left(\frac{M_\eta^2}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{y_{N,i_2,r_1}} \overline{\lambda_5} \overline{\ell_{a_3,i_1}}^T C P_L \ell_{a_2,i_2} H_{a_1} H_{a_4} \overline{\ell_{a_1,a_2}} \overline{\ell_{a_3,a_4}} M_N^3}{8 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{h \overline{\ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T \overline{y_{N,i_1,r_1}} \overline{y_{N,i_2,r_1}} \overline{\ell_{a_1,a_3}} \overline{\ell_{a_4,a_2}} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{8 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{h \log \left(\frac{M_\eta^2}{M_N^2} \right) \overline{\ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T \overline{y_{N,i_1,r_1}} \overline{y_{N,i_2,r_1}} \overline{\ell_{a_1,a_3}} \overline{\ell_{a_4,a_2}} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{8 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{h \overline{y_{N,i_1,r_1}} \overline{y_{N,i_2,r_1}} \overline{\lambda_5} \overline{\ell_{a_3,i_1}}^T C P_L \ell_{a_2,i_2} H_{a_1} H_{a_4} M_\eta^2 \overline{\ell_{a_1,a_2}} \overline{\ell_{a_3,a_4}} M_N}{8 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{h \overline{\ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T M_\eta^2 \overline{y_{N,i_1,r_1}} \overline{y_{N,i_2,r_1}} \overline{\ell_{a_1,a_3}} \overline{\ell_{a_4,a_2}} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N}{8 \left(M_\eta^2 - M_N^2 \right)^2}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_N^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^4}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^4}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_N^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& + \frac{3h \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_N^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& + \frac{3h \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{3h \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& + \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_\eta^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{3h \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_\eta^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{3h \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& + \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_\eta^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{3h \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^4 y_{N,i_3,r_1} H_{a_1}^\dagger}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{baf}2}}{M_\eta^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^4 y_{N,i_3,r_1} H_{a_1}^\dagger}{16 \left(M_\eta^2 - M_N^2 \right)^2}
\end{aligned}$$

$$3 \quad \frac{1}{2} \quad 2 \quad 0$$

$$\begin{aligned}
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^4}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^4}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& + \frac{3h \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& + \frac{3h \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{3h \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& + \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{3h \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& + \frac{3h \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger}{8 \left(M_\eta^2 - M_N^2 \right)} \\
& - \frac{3h \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^4 y_{N,i_3,r_1} H_{a_1}^\dagger}{16 \left(M_\eta^2 - M_N^2 \right)^2} \\
& - \frac{3h \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^4 y_{N,i_3,r_1} H_{a_1}^\dagger}{16 \left(M_\eta^2 - M_N^2 \right)^2}
\end{aligned}$$